

RESEARCH INTERESTS

I am broadly interested in Computational Linguistics and NLP, specifically human-AI alignment, developing robust evaluation frameworks for LLMs, and investigating model internals to advance LLM interpretability and AI safety.

EDUCATION

Jahangirnagar University

Master of Science in Computer Science

Dhaka, Bangladesh

May 2025 – Sept. 2026

- **CGPA:** 3.98/4.00
- **Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Natural Language Processing (A+), Data Mining (A), Artificial Intelligence & Expert Systems (A+), Computational Intelligence (A+)
- **CS Prerequisites:** Data Structures & Algorithms (A+), Software Engineering & Database Systems (A+), Computer Architecture & Operating Systems (A+), Programming Languages (A+)

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Civil Engineering, with Honors

Dhaka, Bangladesh

Feb. 2020 – Mar. 2025

- **CGPA:** 3.77/4.00, **Major:** Transportation Engineering, **Minor:** Environmental Engineering
- **Math and Programming Coursework:** Differential Equations & Statistics (A+), Differential, Integral Calculus and Matrices, Computer Programming Sessional (A+), Engineering Computation Sessional (A+)

RESEARCH EXPERIENCE

Graduate Researcher

Computational Intelligence, Data and Language Research Lab

Supervisor: Dr. Md. Musfique Anwar

Jun. 2025 – Present

Jahangirnagar University

- * **Mechanistic Interpretability:** Co-conceptualized a study of arithmetic heuristic neurons across three Llama models (1B/3B/8B), testing (1) whether arithmetic neurons are form-invariant across symbolic, python code, and natural language word-problem formulations and (2) whether cross-format failures reflect distinct circuits or differential activation of a shared circuit; conducted experiments on the 1B-parameter model and co-authored the manuscript. Identified a compact, causally sufficient shared circuit whose activation transfer recovered over 97% of cross-format failures for addition and subtraction, with lower recovery for multiplication and division.
- * **Event Extraction:** Led a comprehensive study of event detection from Bangla news, developing a generalized news event ontology, coordinating annotation of 9,979 sentences across 40 subtypes and three noise conditions, and training and evaluating encoder-based and decoder-only models; found decoder-only LLMs markedly more robust to noise, while annotation-guided instruction tuning improved performance but did not consistently improve robustness.
- * **Cultural Alignment:** Led *BanglaSocialBench* end-to-end, integrating LLM APIs, building evaluation pipelines, coordinating annotation, and conducting statistical analyses for a 1,719-instance benchmark evaluated across 12 LLMs; uncovered systematic over-politeness, failure to recognize multiple socially acceptable address-pronoun forms, and cross-religious kinship-term conflation (ACL SRW 2026, Best Paper Award).

Research Intern

Maks Inc.

May 2025 – Oct. 2025

Dhaka, Bangladesh (HQ: Frisco, TX, USA)

- * Conducted literature reviews and collaborated with data scientists and transportation engineers to analyze large-scale BRTA traffic datasets, develop emission models, perform geospatial analyses, and conduct in-person stakeholder interviews across major administrative divisions of Bangladesh; prepared technical reports for projects funded by the **World Bank** and **Asian Development Bank**.

Undergraduate Researcher

Advised by: Dr. Md. Shamsul Hoque & Dr. Md Asif Raihan

Bangladesh University of Engineering and Technology

Nov. 2023 – Feb. 2025

- * Built an end-to-end computer vision and machine learning framework integrating street-view imagery, open geospatial data, and crash records to determine the impact of visual and built-environment characteristics on pedestrian crash severity, applying SHAP-based interpretability techniques.

PUBLICATIONS & PREPRINTS

* indicates equal contribution

- [1] Sharath Naganna*, Tanvir Ahmed Sijan*, Uddipta Kalita. **"Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs."** *arXiv preprint arXiv:2607.16693*, July, 2026. [\[arXiv\]](#)
- [2] Tanvir Ahmed Sijan, S. M Golam Rifat, Nayeemul Islam, Md. Musfique Anwar. **"Beyond Clean Text: Evaluating Encoder and Decoder Robustness for Bangla Event Detection in Noisy Text."** *Accepted at the 11th Workshop on Natural User-generated Text (W-NUT), co-located with EMNLP 2026*, August, 2026. [\[arXiv\]](#)
- [3] Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar. **"BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction."** *ACL Student Research Workshop (Oral)*, 2026. **[Best Paper Award]** [\[ACL Anthology\]](#)
- [4] Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque. **"Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data."** *Submitted to Transportation Research Board*, 2026.

HONORS & AWARDS

Best Paper Award <i>ACL Student Research Workshop 2026</i>	Jul. 2026
Degree Awarded with Honors <i>Bangladesh University of Engineering and Technology (BUET)</i>	Mar. 2025
Undergraduate Student Research Grant <i>Research and Innovation Centre for Science and Engineering, BUET</i> ◦ Awarded competitive student research grant for developing a deep learning based framework, incorporating street-view imagery and open geospatial data with SHAP-based interpretability, to model pedestrian-vehicle crash severity.	Jan. 2024
Dean's List Scholarship <i>Bangladesh University of Engineering and Technology (BUET)</i>	Jul. 2019
Higher Secondary Certificate (HSC) Scholarship <i>Board of Intermediate and Secondary Education, Rajshahi</i> ◦ Awarded Talentpool Scholarship; ranked 10th (~top 0.01%) among all candidates in the Rajshahi Board.	2019

SERVICE AND LEADERSHIP

Workshop Reviewer <i>11th Workshop on Natural User-generated Text (W-NUT) collocated with EMNLP 2026</i>	Jul. 2026
Deputy Vice President <i>ITE BUET Student Chapter</i> ◦ Collaborated with executive members to organize workshops and seminars on various topics of transportation research.	Dec. 2024 – Mar. 2025
Vice President <i>ACI BUET Student Chapter</i> ◦ Organized poster competitions, seminars, and field visits in coordination with chapter members.	Mar. 2024 – Mar. 2025

SKILLS

Programming: Python, C, C++, SQL, MATLAB
Libraries: PyTorch, NumPy, Pandas, Transformers, Matplotlib, Scikit-Learn
Other: Linux, Git, Bash, $\LaTeX$
Languages: Bangla (Native), English, Hindi (Basic)